

iPSC-EC BO Model Data Summary

iPSC-EC Differentiation Data — Hall/Lin/Ogle (2025) Extraction

Purpose: reference dataset for the Bayesian Optimization model project (iPSC-to-endothelial-cell differentiation environment optimization). This paper was assigned as a real, published factor→outcome dataset to validate the BO approach against — it is not the paper the original task doc’s example factors (FGF, IL-6, temperature) were describing; that was a mismatch in the task doc, unrelated to this paper.

Source: Hall, M.L., Lin, W.-H. & Ogle, B.M. Optimizing extracellular matrix for endothelial differentiation using a design of experiments approach. *Scientific Reports* 15, 24479 (2025). <https://doi.org/10.1038/s41598-025-09256-9>

What the paper studies: a Design of Experiments (DoE) optimization of ECM protein-coating composition (Collagen I, Collagen IV, three laminin subtypes, fibronectin) for driving human iPSC differentiation into endothelial cells, measured by CD31 immunofluorescence (CD31 area / DAPI area, normalized to a fibronectin-only control).

How this data was produced: none of it exists as printed numbers in the paper — it’s all read off box-and-whisker or bar charts. Values were extracted by pulling each figure at native 300 DPI resolution from the source PDF, calibrating pixel position against the axis gridlines, and detecting individual data points (or, where that broke down, reading values directly off zoomed images). Full methodology and per-figure troubleshooting notes are in `EXTRACTION_NOTES.md`, included alongside this doc.

Data reliability — read this before using any number below

Dataset	Method	Confidence
Stage 1 (Table 1 / Fig 1a)	Pixel-calibrated automated dot detection, spot-checked against paper’s own claims	High
Stage 2 (Table 2 / Fig 2a)	Same automated method	High
Stage 3 (Fig 3b)	Automated, with manual removal of statistical-significance-bracket artifacts	Medium
Stage 4a-d (Fig 4a-d)	Visual reading (automated dot-clustering broke down on overlapping points)	Medium-low, ±0.2-0.3 CD31 units
Stage 4e (Fig 4e)	Visual reading, reported as 5-number summary only (min/Q1/median/Q3/max), not individual replicates	Low — coarsest data here
Stage 5 (Fig 5b-d)	Visual bar+error-bar reading; different chart type and units (µm/mm², #/mm², not CD31/DAPI)	Low, except the two CIV% values in the dual-construct chart, which are exact figures quoted directly from the paper’s text

Two things worth flagging to whoever uses this: - The Stage 1 Results text states Collagen IV’s high concentration was 28 µg/mL; the Methods section says 56 µg/mL. We used 56 (and its Stage-2-doubled value, 112) because the paper’s own reported optimum (CIV = 67.2 µg/mL) only fits inside the tested range if the true high is 112 — this resolves the discrepancy but isn’t independently confirmed against raw data. - In Fig 3b, one factor value (LN411 concentration in the “LN411+FN” reference condition) is unknown — it references a concentration from an earlier paper (ref. 28) not given numerically here. Marked `unknown_prior_published_conc` in that table.

Stage 1 — Screening factorial (Table 1 / Fig 1a): 23 conditions, 6 factors

ID	C (µg/mL)	CIV (µg/mL)	LN111 (µg/mL)	LN411 (µg/mL)	LN511 (µg/mL)	FN (µg/mL)	CD31 mean	Variance	n
stage1_01	0	0	0	0	0	22	0.966	0.0943	5
stage1_02	0	0	0	0	0.8	75	0.526	0.0824	4
stage1_03	0	0	0	0.8	0	75	1.665	1.7328	5
stage1_04	0	0	0	0.8	0.8	22	1.001	0.104	2
stage1_05	0	0	15.8	0	0	75	0.723	0.1518	4
stage1_06	0	0	15.8	0.8	0	22	1.41	0.4545	4
stage1_07	0	0	15.8	0.8	0.8	75	0.961	0.4409	4
stage1_08	0	56	0	0	0	75	1.155	0.6892	5
stage1_09	0	56	0	0.8	0	22	1.098	0.1832	3
stage1_10	0	56	0	0.8	0.8	75	0.542	0.2264	4
stage1_11	0	56	15.8	0	0.8	22	0.651	0.167	4
stage1_12	0	56	15.8	0.8	0	75	0.727	0.2914	5
stage1_13 (center)	17.75	28.0	7.9	0.4	0.4	48.5	0.626	0.0584	3
stage1_14	35.5	0	0	0	0	75	1.213	0.3938	5
stage1_15	35.5	0	0	0.8	0	22	0.929	0.0605	3
stage1_16	35.5	0	0	0.8	0.8	75	0.94	0.3243	4
stage1_17	35.5	0	15.8	0	0.8	22	1.21	0.0741	3
stage1_18	35.5	0	15.8	0.8	0	75	0.854	0.1752	5
stage1_19	35.5	56	0	0	0.8	22	0.854	0.0513	3

stage1_20	35.5	56	0	0.8	0	75	1.955	1.9957	5
stage1_21	35.5	56	15.8	0	0	22	0.714	0.0178	3
stage1_22	35.5	56	15.8	0	0.8	75	1.203	0.4766	3
stage1_23	35.5	56	15.8	0.8	0	22	1.406	0.4937	4

Stage 2 — Central composite design (Table 2 / Fig 2a): 25 conditions, 4 factors

ID	C (µg/mL)	CIV (µg/mL)	LN411 (µg/mL)	FN (µg/mL)	CD31 mean	Variance	n
stage2_01	35.5	56.0	0	48.5	0.564	0.013	2
stage2_02	35.5	56.0	0.8	75	2.396	3.212	6
stage2_03	0	0	0	75	0.65	0.1223	4
stage2_04 (center)	35.5	56.0	0.8	48.5	2.957	1.3214	7
stage2_05	0	0	1.6	75	1.691	0.2635	3
stage2_06	71	0	0	75	1.223	0.5457	5
stage2_07	71	112	1.6	75	1.677	0.7661	6
stage2_08	71	0	1.6	22	3.088	6.8805	7
stage2_09	35.5	0	0.8	48.5	2.833	1.4344	9
stage2_10	0	112	0	22	1.686	0.9205	6
stage2_11	0	56.0	0.8	48.5	1.766	0.6426	8
stage2_12	71	56.0	0.8	48.5	2.285	1.1602	8
stage2_13	35.5	56.0	0.8	22	3.667	1.8438	10
stage2_14	71	0	1.6	75	2.176	0.4339	3
stage2_15	71	112	0	22	0.884	0.3168	4
stage2_16	0	112	1.6	22	2.222	0.7758	7
stage2_17	35.5	112	0.8	48.5	3.294	3.3697	6
stage2_18	0	0	1.6	22	3.042	1.9439	6
stage2_19	71	112	1.6	22	3.381	5.53	7
stage2_20	71	0	0	22	0.692	0.054	5
stage2_21	0	112	1.6	75	2.336	0.466	6
stage2_22	35.5	56.0	1.6	48.5	3.06	1.4486	7
stage2_23	71	112	0	75	0.836	0.2238	5
stage2_24	0	112	0	75	0.966	0.1737	8
stage2_25	0	0	0	22	0.846	0.2363	3

Stage 3 — Named validation formulations (Fig 3b): 9 conditions

ID	C (µg/mL)	CIV (µg/mL)	LN411 (µg/mL)	FN (µg/mL)	CD31 mean	Variance	n
TheO	35.6	67.2	0.9	22	0.931	0.1354	5
TheO-C	0	67.2	0.9	22	0.651	0.0295	3
EO	35.6	67.2	0.9	0	6.352	2.3922	5
EO-C	0	67.2	0.9	0	4.924	2.5181	5
++++ (max all)	71	112	1.6	75	0.528	—	1
-+ ++ (max minus C)	0	112	1.6	75	0.873	0.2285	5
+++- (max minus FN)	71	112	1.6	0	2.039	0.834	5
LN411+FN (prior published ref.)	0	0	unknown_prior_published_conc	75	2.097	0.4562	4
FN alone (baseline)	0	0	0	75	0.94	0.1805	4

Stage 4a-d — Single-additive treatments (Fig 4a-d): 10 conditions

ID	Base	SB	VEGF	ECM readded	TGFβ	CD31 mean	Variance	n
fig4a_TheO	TheO	0	0	0	0	0.98	0.1083	5
fig4a_EO	EO	0	0	0	0	5.825	2.1758	4
fig4a_TheO+VEGF	TheO	0	1	0	0	5.38	7.197	5

fig4a_EO+VEGF	EO	0	1	0	0	9.5	25.772	6
fig4b_TheO+ECM	TheO	0	0	1	0	3.5	0.575	5
fig4b_EO+ECM	EO	0	0	1	0	5.825	2.1758	4
fig4c_TheO+SB	TheO	1	0	0	0	3.04	2.263	5
fig4c_EO+SB	EO	1	0	0	0	5.68	7.807	5
fig4d_TheO+TGFβ	TheO	0	0	0	1	0.66	0.113	5
fig4d_EO+TGFβ	EO	0	0	0	1	2.7	1.508	6

Stage 4e — Combinatorial additives (Fig 4e): 10 conditions, 5-number summary

ID	Base	SB	VEGF	ECM	Min	Q1	Median	Q3	Max
TheO_000	TheO	0	0	0	0.5	0.7	0.8	1.0	1.3
TheO_SB+VEGF	TheO	1	1	0	0.8	1.2	1.5	1.9	2.3
TheO_SB+ECM	TheO	1	0	1	0.3	0.5	0.7	1.1	1.6
TheO_VEGF+ECM	TheO	0	1	1	5.8	6.5	8.0	9.5	11.3
TheO_SB+VEGF+ECM	TheO	1	1	1	0.5	0.7	0.8	1.0	1.3
EO_000	EO	0	0	0	3.8	4.5	5.5	7.5	8.8
EO_SB+VEGF	EO	1	1	0	10.5	12.5	15.0	26.0	34.0
EO_SB+ECM	EO	1	0	1	6.0	10.0	15.5	31.0	35.0
EO_VEGF+ECM	EO	0	1	1	9.0	12.0	19.0	24.5	27.0
EO_SB+VEGF+ECM	EO	1	1	1	9.0	13.0	16.0	33.0	41.0

Note: TheO/EO baselines here read visually different from the same-named baselines in Stage 4a-d (e.g. EO median ~5.5 here vs. panels a-d ~6-7). The paper doesn't say whether panel e reuses the same baseline data or reran it independently — treated as separate here rather than assumed identical.

Stage 5 — 3D bioprinted constructs (Fig 5b-d): different units, not directly comparable to Stages 1-4

Chart	Condition	Mean	SD	n	Source
CD31 area (norm. to No ECM Ctrl)	EO	1.85	0.5	6	visual
CD31 area (norm. to No ECM Ctrl)	CO	0.45	0.15	3	visual
CD31 area (norm. to No ECM Ctrl)	CO-2FN	0.25	0.15	3	visual
CD31 area (norm. to No ECM Ctrl)	No ECM Ctrl	1.0	0.4	3	visual
Total vessel length (µm/mm²)	EO	870	230	6	visual
Total vessel length (µm/mm²)	CO	220	100	3	visual
Total vessel length (µm/mm²)	CO-2FN	130	80	3	visual
Total vessel length (µm/mm²)	No ECM Ctrl	470	180	3	visual
Branch points (#/mm²)	EO	6.5	1.8	6	visual
Branch points (#/mm²)	CO	1.0	0.6	3	visual
Branch points (#/mm²)	CO-2FN	0.4	0.3	3	visual
Branch points (#/mm²)	No ECM Ctrl	3.2	1.5	3	visual
CIV area % (dual construct)	EO side	81.29	11.08	4	exact, from paper text
CIV area % (dual construct)	Control side	30.26	12.09	4	exact, from paper text
CD31 area % (dual construct)	EO side	0.63	0.27	4	visual
CD31 area % (dual construct)	Ctrl (CIV–) side	0.18	0.15	4	visual

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This document plus the six underlying CSVs (stage1_... through stage5_...) and EXTRACTION_NOTES.md (full methodology detail) are all in the same output folder — share the whole set if others need to load the CSVs directly into a notebook rather than retype from these tables.